JAVA PRACTICAL PROGRAM REPORT

Farm Animal Inheritance and Method Overriding

Program/Class Name: FarmAnimalExample.java

Language: Java

Topic: Inheritance using extends; method overriding; objects; inherited eat() method; subclass-specific sound() implementations.

This report presents the aim, requirements, program logic, UML-style class diagram, source code, execution commands, sample console output, and conclusion.

1. Aim

To demonstrate inheritance and method overriding using FarmAnimal as the parent class and Cow and Goat as subclasses.

2. Requirements

• Java Development Kit (JDK)
• Text editor or Java IDE
• Save the source code using the public class name as the filename.

3. Concepts Used

Inheritance using extends; method overriding; objects; inherited eat() method; subclass-specific sound() implementations.

4. Algorithm

1. Start the program.
2. Create an object of Cow and an object of Goat.
3. Call eat() on Cow; the inherited parent-class method runs.
4. Call sound() on Cow; the overridden Cow method runs.
5. Call eat() on Goat; the inherited parent-class method runs.
6. Call sound() on Goat; the overridden Goat method runs.
7. End the program.

5. UML / Class Diagram

FarmAnimal (parent class): eat(), sound()
Cow extends FarmAnimal: overrides sound()
Goat extends FarmAnimal: overrides sound()
FarmAnimalExample.main(): creates Cow and Goat objects and calls methods

The diagram summarizes the main class flow and the key methods or calculations used by the program.

6. Source Code

```
class FarmAnimal {

    void eat() {
        System.out.println("Farm animal eats food");
    }

    void sound() {
        System.out.println("Farm animal makes a sound");
    }
}

class Cow extends FarmAnimal {

    void sound() {
        System.out.println("Cow says: Moo");
    }
}

class Goat extends FarmAnimal {

    void sound() {
        System.out.println("Goat says: Meh");
    }
}

public class FarmAnimalExample {
    public static void main(String[] args) {

        Cow c = new Cow();
        Goat g = new Goat();

        c.eat();
        c.sound();

        g.eat();
        g.sound();
    }
}
```

7. Compilation and Execution

```
javac FarmAnimalExample.java
java FarmAnimalExample
```

8. Sample Console Output

The source code was compiled and run in the available Java environment. For interactive programs, typed inputs are shown beside their prompts for readability.

```
Farm animal eats food
Cow says: Moo
Farm animal eats food
Goat says: Meh
```

9. Result / Conclusion

The program demonstrates inheritance because Cow and Goat inherit eat() from FarmAnimal. It demonstrates method overriding because each subclass provides its own implementation of sound().